

```
In [1]: import pandas as pd
df=pd.read_excel("sales_raw_500.xlsx")
df
```

```
Out[1]:
```

	Order_ID	Order_Date	Customer	Region	Product	Sales	Cost
0	1001	2023-01-13	Rohit	South	Mobile	23289	19118
1	1002	2023-10-07	Aman	West	Laptop	8905	5593
2	1003	2023-11-05	Rahul	South	Monitor	59987	39959
3	1004	2024-02-19	Rahul	South	Printer	49597	33892
4	1005	2024-01-26	Anjali	North	Laptop	54797	34468
...	...	...	...	...	...	...	...
515	1389	2024-02-12	Vikas	North	Laptop	17924	12145
516	1496	2023-04-24	Kiran	North	Printer	23434	16388
517	1031	2023-02-19	Vikas	East	Printer	58883	43579
518	1317	2023-07-05	Aman	North	Laptop	76781	66865
519	1409	2024-01-16	Pooja	North	Monitor	19969	13586

520 rows × 7 columns

```
In [5]: #duplicate removing
df=df.drop_duplicates()
df
```

Out[5]:

	Order_ID	Order_Date	Customer	Region	Product	Sales	Cost
0	1001	2023-01-13	Rohit	South	Mobile	23289	19118
1	1002	2023-10-07	Aman	West	Laptop	8905	5593
2	1003	2023-11-05	Rahul	South	Monitor	59987	39959
3	1004	2024-02-19	Rahul	South	Printer	49597	33892
4	1005	2024-01-26	Anjali	North	Laptop	54797	34468
...	...	...	...	...	...	...	...
495	1496	2023-04-24	Kiran	North	Printer	23434	16388
496	1497	2023-06-27	Anjali	South	Laptop	49007	35970
497	1498	2023-05-08	Aman	East	Tablet	19160	15636
498	1499	NaT	Pooja	West	Mobile	59266	44326
499	1500	2023-10-07	Vikas	North	Printer	34567	22426

500 rows × 7 columns

In [8]:

```
# fill the blank
df['Order_Date']=pd.to_datetime(df['Order_Date'],errors='coerce')#to_datetime() is
#other formats into the datetime data type.
#coerce means force a value into a particular data type. If a value cannot be conve
```

In [10]:

```
df=df.dropna(subset=['Order_Date'])
df
```

Out[10]:

	Order_ID	Order_Date	Customer	Region	Product	Sales	Cost
0	1001	2023-01-13	Rohit	South	Mobile	23289	19118
1	1002	2023-10-07	Aman	West	Laptop	8905	5593
2	1003	2023-11-05	Rahul	South	Monitor	59987	39959
3	1004	2024-02-19	Rahul	South	Printer	49597	33892
4	1005	2024-01-26	Anjali	North	Laptop	54797	34468
...	...	...	...	...	...	...	...
494	1495	2023-10-18	Rahul	South	Monitor	46709	40931
495	1496	2023-04-24	Kiran	North	Printer	23434	16388
496	1497	2023-06-27	Anjali	South	Laptop	49007	35970
497	1498	2023-05-08	Aman	East	Tablet	19160	15636
499	1500	2023-10-07	Vikas	North	Printer	34567	22426

482 rows × 7 columns

```
In [13]: df['Profit']=df['Sales']-df['Cost']
df['Year']=df['Order_Date'].dt.year
df['Month']=df['Order_Date'].dt.month
df
```

Out[13]:

	Order_ID	Order_Date	Customer	Region	Product	Sales	Cost	Profit	Year	Mont
0	1001	2023-01-13	Rohit	South	Mobile	23289	19118	4171	2023	
1	1002	2023-10-07	Aman	West	Laptop	8905	5593	3312	2023	1
2	1003	2023-11-05	Rahul	South	Monitor	59987	39959	20028	2023	1
3	1004	2024-02-19	Rahul	South	Printer	49597	33892	15705	2024	
4	1005	2024-01-26	Anjali	North	Laptop	54797	34468	20329	2024	
...	...	...	...	...	...	...	...	...	...	
494	1495	2023-10-18	Rahul	South	Monitor	46709	40931	5778	2023	1
495	1496	2023-04-24	Kiran	North	Printer	23434	16388	7046	2023	
496	1497	2023-06-27	Anjali	South	Laptop	49007	35970	13037	2023	
497	1498	2023-05-08	Aman	East	Tablet	19160	15636	3524	2023	
499	1500	2023-10-07	Vikas	North	Printer	34567	22426	12141	2023	1

482 rows × 10 columns



```
In [14]: df.to_csv('sales_cleaned.csv',index=False)#False it will remove the indexing that i
```

```
In [15]: import os

downloads = os.path.join(os.path.expanduser("~"), "Downloads")

df.to_csv(os.path.join(downloads, "sales_cleaned.csv"), index=False)
```

```
In [16]: #data python+SQL+Excel
import pandas as pd
from sqlalchemy import create_engine
df=pd.read_csv('sales_cleaned.csv')
df
```

```
-----
ModuleNotFoundError                                Traceback (most recent call last)
Cell In[16], line 3
      1 #data python+SQL+Excel
      2 import pandas as pd
----> 3 from sqlalchemy import create_engine
      4 df=pd.read_csv('sales_cleaned.csv')
      5 df

ModuleNotFoundError: No module named 'sqlalchemy'
```

```
In [20]: import sys
print(sys.executable)

C:\Users\USER\miniconda3\python.exe
```

```
In [1]: from sqlalchemy import create_engine
print("SQLAlchemy imported successfully!")

SQLAlchemy imported successfully!
```

```
In [4]: import pandas as pd
from sqlalchemy import create_engine

df = pd.read_csv("sales_cleaned.csv")
df
```

Out[4]:

	Order_ID	Order_Date	Customer	Region	Product	Sales	Cost	Profit	Year	Mont
0	1001	2023-01-13	Rohit	South	Mobile	23289	19118	4171	2023	
1	1002	2023-10-07	Aman	West	Laptop	8905	5593	3312	2023	1
2	1003	2023-11-05	Rahul	South	Monitor	59987	39959	20028	2023	1
3	1004	2024-02-19	Rahul	South	Printer	49597	33892	15705	2024	
4	1005	2024-01-26	Anjali	North	Laptop	54797	34468	20329	2024	
...	...	...	...	...	...	...	...	...	...	
477	1495	2023-10-18	Rahul	South	Monitor	46709	40931	5778	2023	1
478	1496	2023-04-24	Kiran	North	Printer	23434	16388	7046	2023	
479	1497	2023-06-27	Anjali	South	Laptop	49007	35970	13037	2023	
480	1498	2023-05-08	Aman	East	Tablet	19160	15636	3524	2023	
481	1500	2023-10-07	Vikas	North	Printer	34567	22426	12141	2023	1

482 rows × 10 columns



In [9]: `engine = create_engine("mysql+pymysql://root:Tvs%405755@localhost:3306/sales_db")#@`

In [10]: `df.to_sql(name='sales_data', con=engine, if_exists='replace', index=False)`

Out[10]: 482

In []: